

Project 2
CSE 464
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Project Description

Continuing Project 1 and adding 3 new APIs to remove node/nodes and edge and their test cases.

Parse a DOT file to create a directed graph object.

- Add nodes and edges to the graph.
- Output the graph details as text.
- Export the graph in DOT format and generate a graphical output (e.g., PNG) using Graphviz.

Prerequisites

- Java JDK 11 or above
- Maven 3.6+
- Graphviz installed and available in your system PATH
- Git

Instructions to run:

Clone this repo <https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10.git>

Navigate to this project directory : CSE-464-2025-gmehta10

Run this command to compile and test : mvn clean package

Then go to the main file and run it using the run button in IntelliJ app

Feature 1:

<https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10/commit/43762e35a02c445d675dc63560e22ad655e8a386>

Run

GraphParserTest

✓

GraphParserTest

(prc 173 ms)

✓

testParseGraph

22 ms

✓

testOutputDOTGrap

4 ms

✓

testAddNodeAndAd

0 ms

✓

testAddEdge

1 ms

✓

testRemoveNodes

0 ms

✓

testRemoveEdge

1 ms

✓

testRemoveNode

0 ms

✓

testOutputGraphic

140 ms

✓

testRemoveNonExis

5 ms

✓

testRemoveNonExis

0 ms

✓

testRemoveNonExis

0 ms

✓

Tests passed: 11 of 11 tests – 173 ms

/Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home/bin/java ...

Parsed Graph:

Number of nodes: 3

Nodes: [A, B, C]

Number of edges: 2

Edges:

A -> B

B -> C

Process finished with exit code 0



Feature 2:

<https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10/commit/58d397cf2cac64681dde020bfacff0d3e8e0e533>

<https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10/actions/runs/14097604019>

← Java CI with Maven

✓ CI2 #3

Re-run all jobs

⋮

Summary

Jobs

✓ build

Run details

Usage

Workflow file

build

succeeded now in 19s

Search logs

⚙

> ✓ Set up job

1s

> ✓ Checkout repository

1s

> ✓ Set up JDK 11

0s

> ✓ Build with Maven

9s

> ✓ Run tests

4s

✓ Post Set up JDK 11

0s

1 Post job cleanup.

> ✓ Post Checkout repository

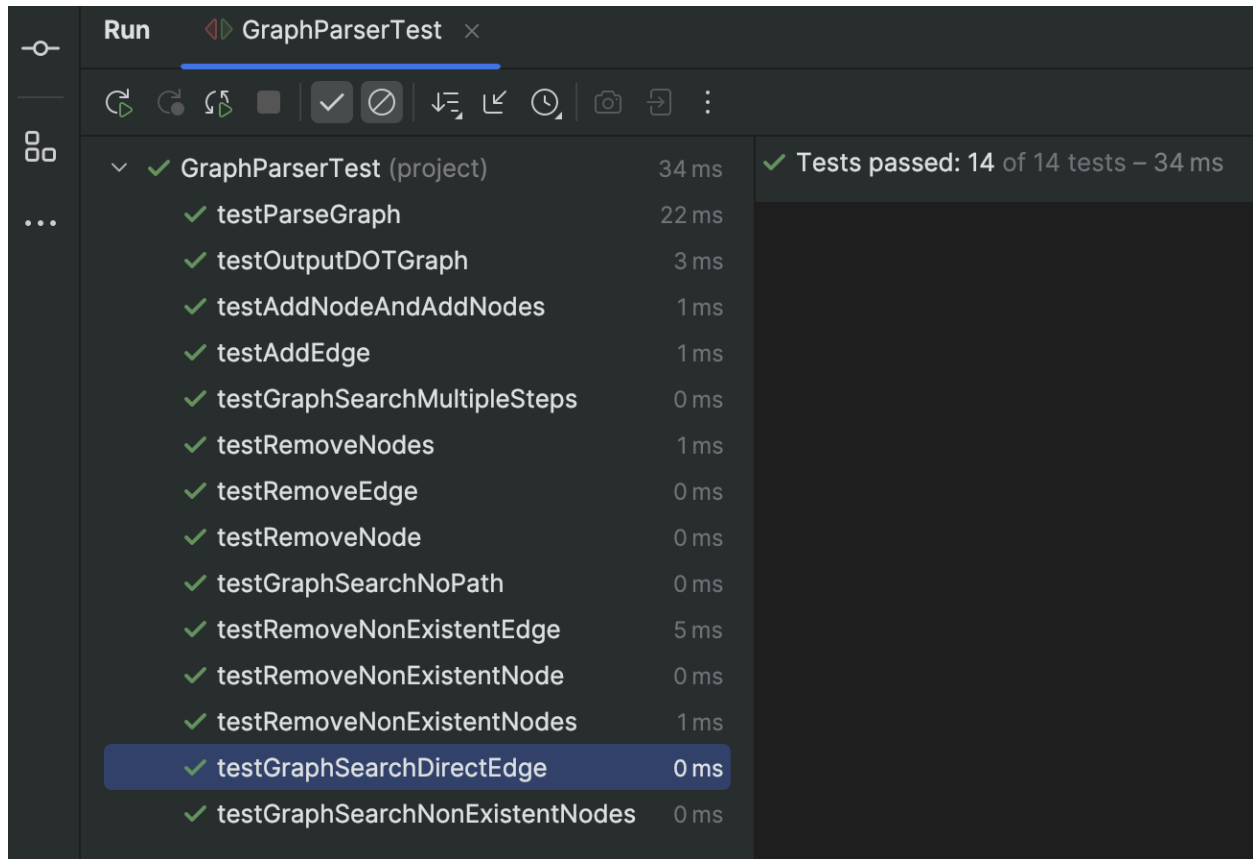
1s

> ✓ Complete job

0s

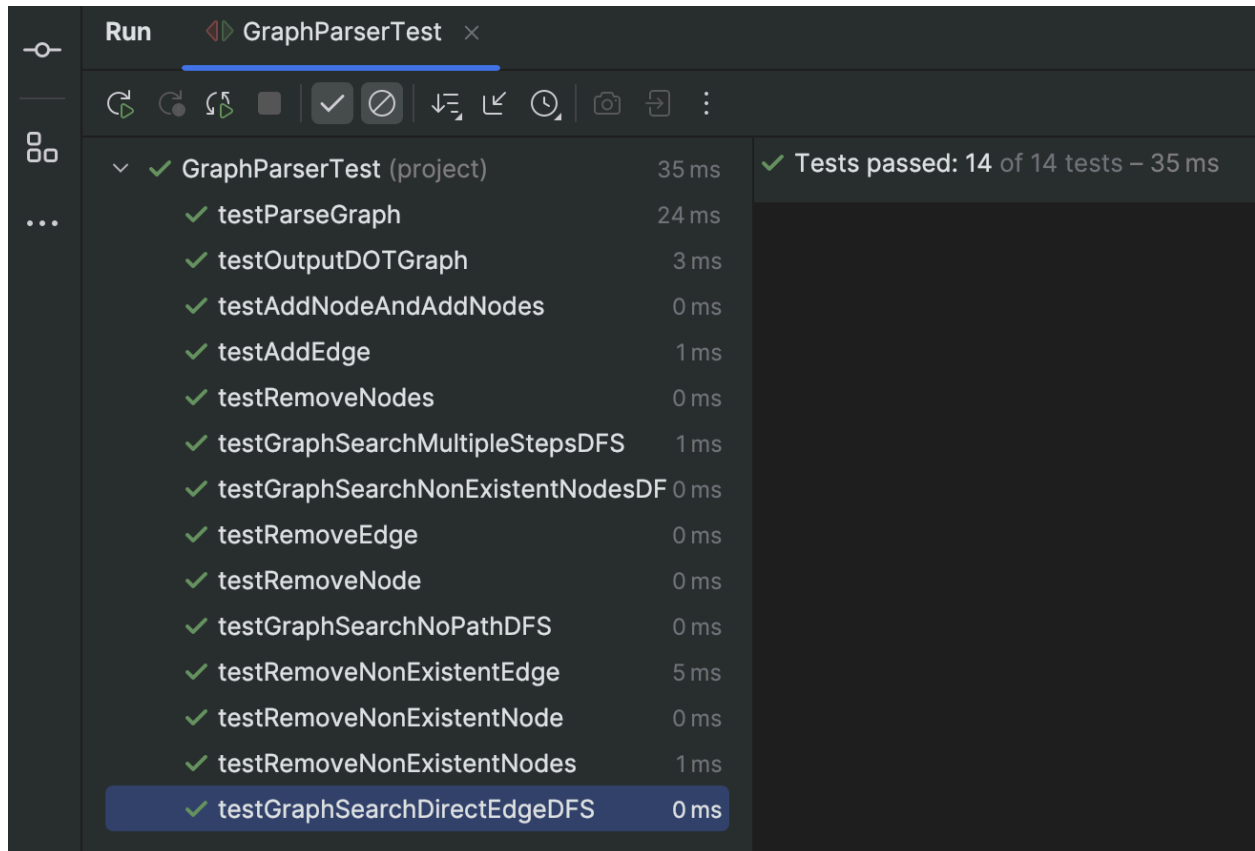
Feature 3:

<https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10/commit/7338aaad24b385dd21474440a78116241c7d641d>



Feature 4:

<https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10/commit/e7e8844a618944de69b22458ebc136dfb0f4d97f>



Feature 5:

<https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10/pull/1> (this is for BFS pull request)

<https://github.com/skidiKidiKaka/CSE-464-2025-gmehta10/pull/2> (this is for DFS pull request)

Run ▶ GraphParserTest x

✓ Tests passed: 18 of 18 tests – 33 ms

✓ GraphParserTest (project)	33 ms
✓ testParseGraph	21 ms
✓ testOutputDOTGraph	3 ms
✓ testAddNodeAndAddNodes	1 ms
✓ testAddEdge	0 ms
✓ testRemoveNodes	0 ms
✓ testUnifiedGraphSearchDirectEdgeBFS	1 ms
✓ testUnifiedGraphSearchDirectEdgeDFS	0 ms
✓ testUnifiedGraphSearchNoPathBFS	1 ms
✓ testUnifiedGraphSearchNoPathDFS	0 ms
✓ testUnifiedGraphSearchNonExistentNodesBFS	0 ms
✓ testUnifiedGraphSearchNonExistentNodesDFS	0 ms
✓ testRemoveEdge	0 ms
✓ testRemoveNode	0 ms
✓ testRemoveNonExistentEdge	5 ms
✓ testRemoveNonExistentNode	0 ms
✓ testRemoveNonExistentNodes	1 ms
✓ testUnifiedGraphSearchMultipleStepsBFS	0 ms
✓ testUnifiedGraphSearchMultipleStepsDFS	0 ms

[← Back to pull request #2](#)

DFS #8

Re-run all jobs

...

Summary

Jobs

 build

Run details

 Usage

 Workflow file

build

succeeded now in 20s



>

✔

Set up job

2s

>

✔

Checkout repository

1s

>

✔

Set up JDK 11

0s

>

✔

Build with Maven

9s

>

✔

Run tests

4s

>

✔

Post Set up JDK 11

0s

>

✔

Post Checkout repository

0s

>

✔

Complete job

0s